**RESEARCH**

My research group is working on advancing the control system technologies and their applications to various engineering systems, such as robotics, mechatronics, and industrial processes, with social impacts and public benefits. The current research focus is on developing new control and optimization frameworks for autonomous robotic vehicle (ARV) systems. The challenges and opportunities of incorporating advanced optimization techniques (e.g., MPC, machine/deep learning) into ARV sensing, navigation, and control system design will be fully investigated. The developed new technology will be applied to enable interesting applications such as vision servoing, autonomous navigation, multi-robot cooperative control, etc.

## Research Interests

- Model Predictive Control Theory and Applications
- Autonomous Underwater Vehicles
- Perception, Navigation, and Control for Robotics
- Vision-Based Intelligent Control Applications
- Cooperative Control of Multi-Agent Systems
- Distributed Control of Large-Scale Systems

## Publications

L ⓘ ao, Z. Yang, Y. Liu, C. Shen, TRCA-Net: Stronger U Structured Network for Human Image Segmentation, Neural Computing and Applications, 2023, In Press, https://doi.org/10.1007/s00521-023-08199-4

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Y. Shi, C. Shen, H. Wei, and K. Zhang, Advanced Model Predictive Control for Autonomous Marine Vehicles, Springer Nature, 2023, In Press, ISBN 978-3-031-19353-8

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Code: https://github.com/ACIPL/MPC4AMVs

N. Yang, C. Shen, M. Johnson-Roberson, and J. Sun, "Energy-optimal Three-dimensional Path-following Control of Autonomous Underwater Vehicles under Ocean Currents", The 2022 IEEE Conference on Control Technology and Applications (CCTA), Trieste, Italy, 2022

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C. Shen, Y. Shi and B. Buckham, “Integrated Path Planning and Tracking Control of an AUV: A Unified Receding Horizon Optimization Approach”, IEEE/ASME Transactions on Mechatronics, vol. 22, no. 3, pp. 1163 - 1173, 2017.

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C. Shen, B. Buckham and Y. Shi, “Modified C/GMRES Algorithm for Fast Nonlinear Model Predictive Tracking Control of an AUV”, IEEE Transactions on Control Systems Technology, vol. 25, no. 5, pp. 1896 - 1904, 2017.

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